

**AI-Powered Mobile Platform Enhancing Public Safety and
Community Policing in Sri Lanka**

25-26J-200

Project Proposal Report
Indudunu I W O

B.Sc. (Hons) Degree in Information Technology specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

July 2025

Safety Prediction for Women & Children

25-26J-200

Project Proposal Report

Indudunu I W O – IT22311054

Supervisor: Mrs. jenny krishara

B.Sc. (Hons) Degree in Information Technology specialized in
Information Technology.

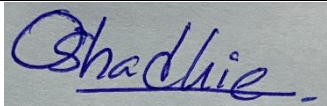
Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

July 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Group Member Name	Student ID	Signature
Indudunu I W O	IT22311054	

Signature of the Supervisor
(Mrs. jenny krishara)

Date

.....

.....

Signature of the Co-Supervisor
(Dr. Dinuka R. Wijendra)

Date

.....

.....

ABSTRACT

Women and children face significant safety issues in Sri Lanka, particularly in situations involving harassment, stalking, domestic violence, child endangerment, and unsafe transport environments. Existing emergency communication platforms and SOS systems are predominantly reactive, relying heavily on victim-initiated voice calls or static alerts, which often fail during high-stress or life-threatening situations. Such systems also lack predictive capabilities, multilingual support, and mechanisms to ensure data reliability, thereby limiting their effectiveness in real-world use. This research proposes the development of an AI-powered Safety Prediction System for Women and Children, designed to filter and classify incoming SOS and incident reports, detect women and child related risks, and generate predictive safety alerts through geospatial clustering and time-series pattern analysis. The system introduces bilingual Natural Language Processing (Sinhala and English) to ensure inclusivity, reducing language barriers that hinder effective reporting. Furthermore, it incorporates a trust-weighted, crowdsourced safety rating board for public transport, institutions, and community areas, safeguarded by AI-based spam filtering to improve the credibility of community-driven safety inputs. By combining incident classification, predictive analytics, crowdsourced ratings, and multilingual guidance delivery, this system shifts from a reactive to a proactive safety model, offering real-time safety maps, heatmaps, and preventive alerts to vulnerable populations. In doing so, it empowers women and children to make informed decisions, enhances community trust, and provides law enforcement with valuable intelligence for better resource allocation. Ultimately, the proposed solution contributes to a citizen-centered, inclusive, and data-driven public safety framework tailored to Sri Lanka's socio-linguistic and cultural context.

Keywords: Safety Prediction, Women and Children, AI Risk Mapping, Crowdsourced Safety Ratings, Sinhala-English NLP, Public Safety, Predictive Alerts

Table of Contents

<i>DECLARATION</i>	2
ABSTRACT	II
<i>LIST OF TABLES</i>	IV
<i>1. INTRODUCTION</i>	1
1.1 Background & Literature survey	3
1.2 Research Gap	8
1.3 Research Problem	11
<i>2. OBJECTIVES</i>	12
2.1 Main Objectives.....	12
2.2 Specific Objective.....	13
<i>3. METHODOLOGY</i>	14
3.1.1 Report Filtering and Classification	14
3.1.2 Risk Mapping and Prediction	14
3.1.3 Crowdsourced Safety Ratings.....	15
3.1.4 Alert and Guidance Delivery	15
3.2 System Architecture Diagram	16
3.3 Project Requirements	18
3.3.2 User Requirements	19
3.3.3 System Requirements	19
3.3.4 Personnel Requirements	19
<i>4. GANTT CHART</i>	20
<i>5. BUDGET AND BUDGET JUSTIFICATION</i>	20
<i>6. COMMERCIALIZATION</i>	21
<i>APPENDICES</i>	23

LIST OF TABLES

Table 1. 1: Comparison between existing systems	9
Table 6. 1: Budget Plan	20

1. INTRODUCTION

Ensuring the safety of women and children has become a pressing social and technological challenge, particularly in developing regions such as Sri Lanka, where incidents of harassment, domestic violence, stalking, and child endangerment often go underreported or inadequately addressed. Despite various initiatives, existing safety infrastructures are largely inadequate in responding to the dynamic and complex nature of modern safety challenges. Traditional emergency response systems such as voice-based police hotlines or mobile SOS applications are predominantly reactive, providing limited situational awareness and relying heavily on the victim's ability to communicate effectively under stress. These approaches often fail during high-pressure situations, resulting in delays, miscommunication, and poor response outcomes.

For instance, Sri Lanka's 119 police hotline continues to serve as the primary emergency communication channel. However, such voice-driven platforms suffer from major drawbacks. Victims may be too distressed to articulate their situation clearly, they may be unable to describe their location, or they may be silenced by perpetrators in sensitive cases such as domestic violence. This creates a critical gap between the need for rapid, accurate emergency reporting and the ability of victims to provide such information. Even where reports are made successfully, they often lack crucial context (e.g., severity of incidents, type of threat, time-specific risk patterns), which hinders the prioritization and allocation of police resources.

Globally, researchers and technologists have begun to explore artificial intelligence (AI), natural language processing (NLP), and geospatial data analysis as tools to create proactive safety systems. Ajmire & Chaurasia highlight how AI-driven crowd-sensing and NLP pipelines can empower women's safety platforms by analyzing real-time reports and identifying threats beyond simple emergency alerts [1].

In the context of geospatial safety mapping, Ponnusamy *et al.* propose combining NLP and location-based clustering to generate risk maps that visualize unsafe areas for women, allowing authorities and citizens to take preventive action [2]. Related work by Jain *et al.* shows that spatiotemporal clustering techniques can predict crime hotspots and time-sensitive risk patterns, which forms the foundation for predictive safety alerts and heatmaps

[8]. These predictive approaches represent a significant improvement over static, reactive safety apps that only respond once an incident has occurred.

Another important aspect of modern safety systems is fairness, inclusivity, and accessibility. Yu emphasizes the algorithmic divide in AI-based safety tools, where systems often neglect marginalized groups due to language or demographic bias [3]. This is particularly relevant in Sri Lanka, where communication occurs in both Sinhala and English, and victims may switch between the two languages in a single report. Research by Perera *et al.* demonstrates the feasibility of building Sinhala-English code-mixed NLP models, providing a technical foundation for bilingual safety systems that minimize accessibility barriers [8].

Parallel to AI-based detection, crowdsourcing mechanisms are increasingly being adopted to build community-driven safety boards. However, such systems are vulnerable to spam, fake input, or malicious manipulation. Rathi & Mishra propose trust-weighted safety rating models that filter unreliable reports, ensuring data credibility and making community-driven safety ratings more reliable [7]. This aligns with the proposed system’s novelty of introducing a crowdsourced safety rating board for streets, transport services, and institutions, supported by AI-based spam filtering and reputation scoring.

Additionally, recent developments in IoT-enabled and mobile-based safety devices show the potential of integrating real-time data collection with AI prediction models. Thakare & Shinde demonstrated how mobile sensors, combined with AI, can predict unsafe events and assist in emergency response planning [6]. Similarly, Kanimozhi & Arputharaj developed a deep learning-based surveillance pipeline to detect threats against women and children, showing how multimodal data (video, audio, text) can strengthen detection accuracy [5]. These studies confirm that safety prediction is achievable when multiple data source reports, geolocation, and community ratings are integrated into a unified AI model.

The proposed research addresses these shortcomings by developing an AI-powered, predictive safety platform specifically tailored to the needs of women and children in Sri Lanka. By integrating bilingual NLP, geospatial clustering, time-series prediction, and trust-weighted safety ratings, this solution goes beyond conventional SOS systems to build a preventive, inclusive, and community-driven safety ecosystem.

Finally, this research proposal emphasizes the importance of new technology in transforming emergency communication and response. The suggested system is a significant improvement over existing SOS methods since it combines deep learning, computer vision, and audio identification into a user-friendly smartphone application. Just as automated detection technologies are altering agriculture and healthcare, AI-powered emergency systems are ready to reshape public safety. This adaptable and intelligent SOS module not only helps people in crisis, but it also improves the operational performance of law enforcement authorities. The system lays the way for a safer society by responding faster, smarter, and more evidence-based, and it serves as a model for future emergency communication frameworks in Sri Lanka and elsewhere.

1.1 Background & Literature survey

To ground this research in real-world perspectives, a survey was conducted with 10 participants in Sri Lanka. The survey targeted young adults and produced important insights into perceptions of safety, reporting behavior, and expectations of technology-enabled safety systems.

As shown in Figure 1.1, most participants reported low confidence in their safety. Half (50%) stated they feel Unsafe in their daily environment, 40% felt Neutral, while only 10% felt Safe. None reported feeling Very Safe.

This outcome illustrates the mistrust in current systems such as police hotlines or CCTV monitoring. Similar findings were noted by Ajmire & Chaurasia who observed that women often do not feel reassured by reactive SOS applications that rely on manual reporting [1]. The neutrality of 40% suggests resignation — users neither trust nor fully distrust the system but remain unconvinced of its effectiveness. This underlines the need for predictive and preventive features that restore confidence through visible, data-driven safety alerts.

4. How safe do you feel in your daily environment?

10 responses

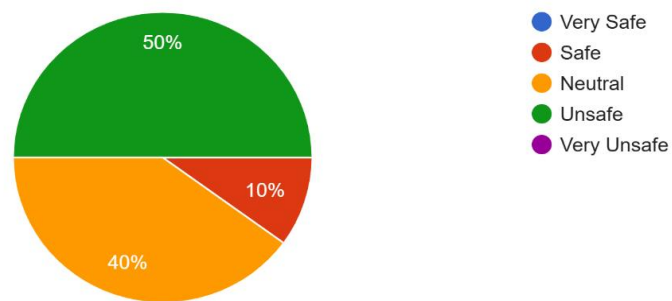


Figure 1.1: Safety perception

Figure 1.2 shows that 70% of respondents had faced or knew someone who faced a safety incident, while only 30% said no. Among those who answered yes, unsafe public transport emerged as the most common threat (75%), followed by street harassment (25%).

This is consistent with Goyal & Thakur who identified unsafe public transport — overcrowding, lack of surveillance, and harassment during commuting — as one of the primary barriers to women’s mobility in South Asia [16]. For Sri Lanka, where buses and trains are the most widely used modes of transport, this insight suggests that transport corridors must be prioritized in predictive safety mapping and rating systems.

6. If yes, what type of issue was most common?

8 responses

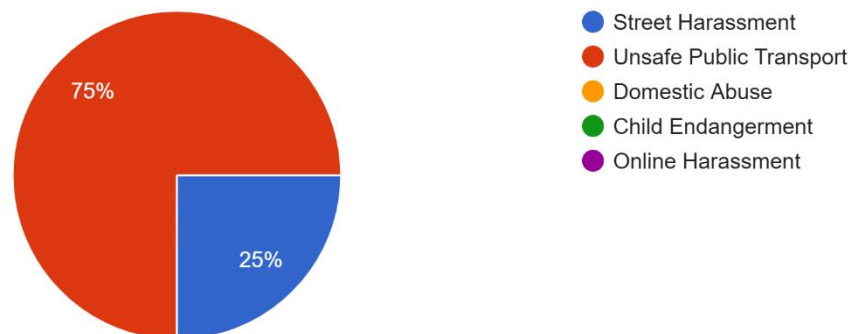


Figure 1.2: Type of issue

Contrary to popular assumptions that risk is highest at night, participants identified multiple unsafe time windows (Fig. 1.3). 40% highlighted morning hours (6 AM – 12 PM), 30% afternoon (12 PM – 6 PM), and 30% evening/night (6 PM – 12 AM). None reported early morning (12 AM – 6 AM) as a key risk period.

This finding is significant: risks are not restricted to “after dark” but extend into commuting and working hours. Kumar & Kaur stress that spatiotemporal clustering models should account for diverse time frames, as crime and harassment often occur when public spaces are crowded, not necessarily deserted [10]. For system design, this indicates that predictive alerts must be dynamic, considering both time and location.

7. At what time do you think women and children are most vulnerable?

10 responses

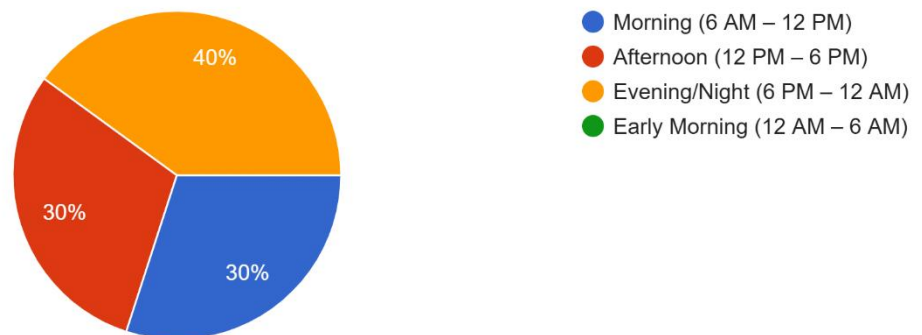


Figure 1.3: Time of vulnerability

Participants were asked to identify the main causes behind safety problems (Fig. 1.4). The majority (50%) attributed incidents to alcohol/drug influence, while 20% pointed to lack of police presence, 20% to poor awareness of laws, and 10% to negligence.

This mix of behavioral and structural factors mirrors findings by Sarma & Prakash who observed that harassment and violence in urban areas are often triggered by alcohol consumption and amplified by weak enforcement [11]. The emphasis on policing also connects to Ahmed *et al.* who found that low trust in law enforcement systems discourages citizens from reporting issues [13]. For the proposed solution, this reinforces the need to augment traditional policing with AI-powered predictive insights and community validation, thereby compensating for systemic weaknesses.

8. In your opinion, what are the main reasons safety issues occur?

10 responses

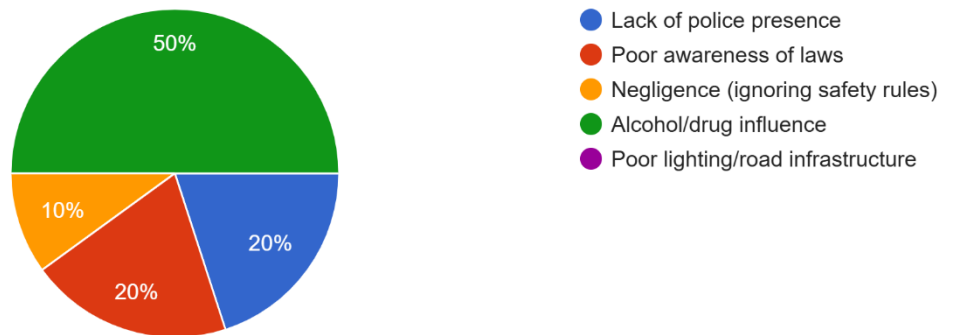


Figure 1.4: Causes of safety issues

When asked about features for a mobile safety system, the vast majority (80%) selected automatic AI-based detection of women/child-related incidents, while 20% favored crowdsourced safety ratings (Fig. 1.5).

Crucially, 90% of respondents agreed that predictive safety alerts would reduce risks, while 100% rated AI prediction as Important or Very Important. The top risks they wanted prioritized were child abduction/endangerment (50%), followed by harassment & stalking (30%), and unsafe transport (20%).

These insights strongly validate the novelty of this research. While many global safety apps emphasize post-incident SOS triggers, Sri Lankan users expressed demand for intelligent, proactive, and child-inclusive features. While Joshi & Mishra show the feasibility of using ML models to classify child safety risks [18]. Together, these findings demonstrate the alignment between user needs and emerging research trends.

13. Which features would you find most useful in such an app?

5 responses

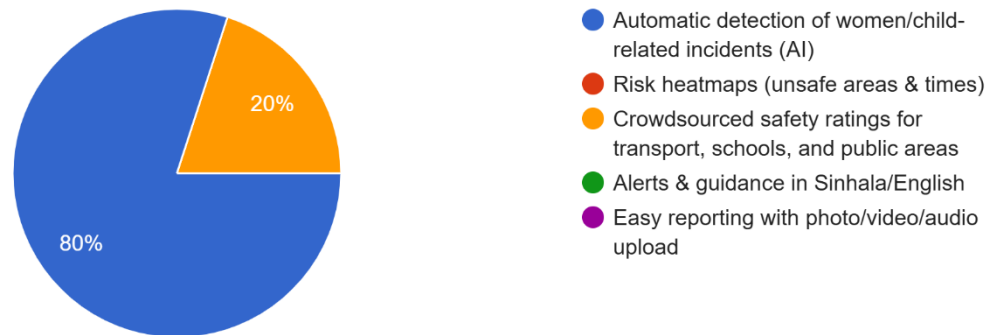


Figure 1.5: Preferred features & predictive alerts

The survey results clearly highlight that existing safety mechanisms in Sri Lanka are not trusted or sufficient. Most participants expressed feeling Unsafe or Neutral in daily environments, with unsafe public transport and harassment identified as the most common threats. Unlike traditional assumptions, risks were also reported during daytime commuting hours, reinforcing the need for time-aware predictive models rather than static night-focused alerts. Alcohol and drug influence was seen as the leading driver of unsafe conditions, reflecting the importance of context-sensitive and community-informed interventions.

Most importantly, the survey confirmed that citizens are willing to adopt digital safety tools if they offer AI-powered intelligence and predictive alerts rather than simple reporting functions. With 90% agreeing that predictive alerts can reduce risks, and child abduction/endangerment ranked as the top priority, the evidence strongly supports the development of an AI-powered, bilingual, predictive safety system. Such a system must combine automatic incident detection, spatiotemporal risk mapping, and trust-weighted safety ratings to provide proactive, community-centered safety guidance for women and children in Sri Lanka.

1.2 Research Gap

Although global research demonstrates promising applications of artificial intelligence in women's and children's safety, significant gaps remain, particularly in the Sri Lankan context. Most existing SOS and reporting applications are still reactive in nature, transmitting only a location pin or short message after an incident occurs. These systems lack predictive intelligence that could warn users before danger arises, resulting in missed opportunities for prevention. Another key limitation is bilingual accessibility. Many international platforms assume monolingual English input, whereas in Sri Lanka, users frequently communicate in Sinhala-English code-mixed language. Current solutions are therefore unable to process such inputs effectively, leaving large user groups excluded from digital safety ecosystems [8], [15].

Crowdsourced reporting has been explored as a way of enhancing community participation in safety systems, but most applications lack trust-weighted filtering mechanisms. Without verification, these platforms remain vulnerable to spam, false reports, and malicious manipulation, undermining their reliability [7], [13]. Additionally, child-specific safety features are underdeveloped. While women's safety is a central focus in many studies, very few systems explicitly address child endangerment or abduction, despite their increasing relevance in both urban and rural communities [18].

Another major gap is the lack of context-aware alerts. Existing safety apps typically provide static warnings or location-based updates, but they fail to integrate time-of-day patterns with geospatial data. Recent research shows that safety risks can peak during unexpected hours such as morning and afternoon, particularly in public transport scenarios [10]. This oversight prevents users from receiving timely, situationally relevant guidance. Finally, many systems in both global and local contexts operate independently of law enforcement agencies. As a result, even when reports are made, they are not seamlessly actionable by police authorities, limiting the overall effectiveness of the intervention chain.

Collectively, these gaps highlight the urgent need for a holistic, predictive, and context-sensitive safety solution tailored to Sri Lanka. Such a system must move beyond reactive reporting and integrate bilingual NLP, spatiotemporal prediction, child-focused features, trust-enabled crowdsourcing, and law enforcement connectivity to create a preventive and inclusive safety ecosystem for women and children.

Table 1. 1: Comparison between existing systems

Application Reference	Live Video/Audio Streaming	AI Analysis (NLP)	AI Analysis (Clustering)	Predictive Alerts	Trust-weighted Crowdsourcing	Multilingual Support (Sinhala/English)
Research A	✗	✓	✗	✗	✗	✗
Research B	✓	✓	✗	✗	✗	✗
Research C	✓	✗	✗	✗	✗	✗
Proposed System	✓	✓	✓	✓	✓	✓

When evaluating the state of technology in women’s and children’s safety applications, three main categories of existing approaches emerge: NLP-based systems, multimodal surveillance pipelines, and wearable/location-prediction devices. While each of these represents progress, none of them fully address the diverse and context-specific needs of Sri Lanka, where bilingual communication, trust in crowdsourced data, and predictive intelligence are critical.

Research A emphasizes the importance of AI and NLP-driven classification pipelines for enhancing women’s safety [1]. Their approach processes incident reports and crowd-sensed data to automatically detect whether an input relates to safety issues such as harassment, stalking, or abuse. This contribution demonstrates that NLP can reduce manual workload for moderators and provide real-time filtering of safety-related text. However, the system is fundamentally reactive: it classifies incidents after they occur without forecasting where or when risks may emerge. It also does not incorporate geospatial clustering or time-of-day analysis, which are critical for predictive mapping. Moreover, while effective in English-only contexts, it does not handle Sinhala-English code-mixing, leaving out a large segment of Sri Lankan users. Finally, it does not address the trustworthiness of inputs, leaving it vulnerable to spam or false alerts. Proposed system builds on these ideas but extends them by integrating bilingual NLP (Sinhala and English) [8], [15], predictive spatiotemporal

clustering [9], [10], and trust-weighted crowdsourced validation [7], [13]. Thus, while Ajmire & Chaurasia prove NLP's potential, your work advances it toward a preventive, inclusive, and community-driven platform.

Research B developed a smart surveillance system for women and child safety using deep learning techniques [5]. Their pipeline fused visual (CCTV/video) and textual/NLP data to detect threats in real time. By combining different data modalities, their system reduces reliance on a single input source, which is a major limitation in many applications. It demonstrated that multimodal AI could detect suspicious behaviors and potentially improve situational awareness for authorities. Despite these contributions, the system remained heavily dependent on surveillance cameras, which are not universally available in Sri Lankan urban or rural areas. Its detection was also even triggered, meaning it could only act once an incident began. It lacked the ability to provide early warning alerts based on historical trends or user-reported risk patterns. Additionally, while innovative, it did not include crowdsourced reporting mechanisms, which are essential in contexts where state infrastructure may be limited. Proposed system complements this by not only enabling multimodal inputs (text, audio, optional media) but also shifting the emphasis from reactive detection to proactive prediction. Using clustering techniques [9], [10], it provides heatmaps of risk zones and issues time-sensitive alerts. Furthermore, by incorporating crowdsourced safety ratings with trust-weighted validation [7], [13], system ensures reliability while empowering communities to play an active role in safety monitoring.

Research C proposed AI-based smart wearables integrated with location prediction to improve women's safety [16]. These devices monitored users' movement in real time and provided alerts when abnormal or risky patterns were detected. Wearables have the advantage of being always with the user, allowing rapid triggering of distress signals without requiring phone access. This line of research showed how IoT and AI can be combined to create highly personalized safety solutions. However, such systems face multiple challenges: they are hardware-dependent (users must buy and wear specific devices), which raises adoption barriers in cost-sensitive contexts like Sri Lanka. They are also individual-centric, protecting the person wearing the device but not contributing to a collective safety map for communities. In addition, they rarely incorporate child-specific protection features such as abduction/endangerment detection [18], nor do they integrate

community-sourced ratings to warn others of unsafe environments. Like earlier systems, they also fail to provide bilingual accessibility. Proposed system avoids these limitations by leveraging mobile apps (more scalable than wearables) and emphasizing community-level safety mapping rather than only personal monitoring. Unlike wearable-only systems, it integrates child-focused risk detection [18], supports Sinhala-English NLP [8], [15], and enables predictive safety alerts that help prevent incidents before they occur.

The proposed system consolidates the strengths of these prior works while addressing their weaknesses. It integrates Sinhala-English NLP to process reports inclusively [8], [15], employs spatiotemporal clustering to generate predictive safety alerts [9], [10], introduces a trust-weighted crowdsourced rating board [7], [13], and incorporates child-specific features [18]. This holistic approach ensures that safety is not only monitored but predicted and prevented, moving from a reactive paradigm to a proactive, community-driven, and inclusive safety ecosystem.

1.3 Research Problem

Despite the growing body of research on AI applications for women’s and children’s safety, existing solutions remain largely reactive and fragmented, particularly in the Sri Lankan context. Current SOS and reporting systems focus on transmitting minimal details such as location or a short message after an incident has already occurred, without providing predictive intelligence to warn users beforehand. As Ajmire & Chaurasia argue, NLP-based systems are effective for classification but fail to deliver preventive alerts that could mitigate risks before escalation [1]. Moreover, most safety applications are developed for monolingual English-speaking contexts, leaving significant populations excluded in countries like Sri Lanka where communication frequently occurs in Sinhala-English code-mixed language [8], [15]. This creates an accessibility gap that undermines inclusivity and user adoption.

Another pressing challenge lies in the reliability of crowdsourced data. While community reporting has been proposed to strengthen safety ecosystems, prior studies highlight that without trust-weight filtering, these systems are vulnerable to false or malicious inputs, reducing their credibility [7], [13]. Additionally, most existing safety technologies are

designed primarily for women, with very few offering targeted features for children, such as abduction or endangerment detection, despite evidence that predictive algorithms can effectively classify child-related risks [19]. Further, as Kumar & Kaur demonstrate, safety risks often follow spatiotemporal patterns, peaking during unexpected hours such as daytime commuting, yet most applications fail to incorporate time-aware clustering and prediction [10]. Finally, current systems frequently operate in isolation from law enforcement agencies, limiting the actionability of reported data and weakening trust in formal safety mechanisms [13].

Therefore, the research problem addressed in this study is the absence of an intelligent, multimodal, and predictive safety system tailored for women and children in Sri Lanka. Existing platforms lack the integration of bilingual NLP, spatiotemporal prediction, child-focused features, and trust-enabled crowdsourcing that are essential for a comprehensive, preventive, and community-driven approach. Without such a system, safety risks will remain underreported, poorly predicted, and ineffectively addressed, perpetuating the vulnerabilities of women and children in everyday environments.

2. OBJECTIVES

The proposed research aims to design and implement a comprehensive, AI-powered safety prediction system tailored for women and children in Sri Lanka. Unlike existing reactive systems, the proposed solution integrates predictive intelligence, bilingual natural language processing, spatiotemporal clustering, and crowdsourced validation mechanisms to proactively identify and mitigate safety risks. By bridging the gaps identified in the literature and survey results, the system seeks to empower vulnerable groups, increase trust in safety reporting, and provide actionable intelligence to both users and law enforcement.

2.1 Main Objectives

The main objective of this research is to develop an AI-powered safety prediction system that proactively identifies, predicts, and communicates risks related to women and children in Sri Lanka, using bilingual alerts, geospatial risk mapping, and crowdsourced trust-weighted safety ratings.

2.2 Specific Objective

- Filter and classify incoming SOS and incident reports to detect women/child-related risks.
- Automatically categorize incidents into relevant classes such as harassment, stalking, domestic abuse, child endangerment, unsafe public transport, and school vicinity risks.
- Predict high-risk zones and times using clustering and time-series analysis.
- Apply geospatial clustering methods to identify safety hotspots from historical and real-time reports.
- Use time-series analysis to capture risk variations by time of day.
- Provide heatmaps and proactive safety alerts.
- Generate real-time push notifications to alert users when entering or approaching high-risk zones (e.g., “High reports of harassment near this bus stop after 6 PM”).
- Enable crowdsourced safety ratings with AI-based spam filtering.
- Introduce a Safety Rating Board where users can evaluate public transport, schools, and community areas for safety.
- Deliver multi-language alerts and safety guidance to users.
- Provide safety notifications, guidance, and awareness tips in both Sinhala and English, increasing inclusivity and adoption.
- Messages may include preventive advice (e.g., “avoid poorly lit routes after 6 PM”), child-safety reminders, or emergency contacts.

3. METHODOLOGY

3.1 Understanding the key pillars of the research domain.

3.1.1 Report Filtering and Classification

Accurate The first stage of the methodology involves developing an AI pipeline that automatically filters and classifies incoming SOS signals and incident reports. These reports may include text, voice, or media inputs submitted through the mobile application. Using NLP combined with media analysis, the pipeline will categorize reports into specific safety-related classes such as:

- Harassment and stalking
- Unsafe public transport
- Domestic abuse
- Child endangerment or abduction
- School vicinity risks

This categorization ensures that reports related to women and children are prioritized and processed separately from unrelated complaints. Previous research by Ajmire & Chaurasia confirmed that NLP-driven classification models can reduce manual screening time and highlight relevant safety issues [1]. In this system, the AI pipeline not only classifies incidents but also acts as a filter against irrelevant or spam submissions, thereby improving the efficiency of downstream risk analysis.

3.1.2 Risk Mapping and Prediction

The predictive component of the methodology involves geospatial clustering and time-series analysis to generate safety heatmaps and forecast high-risk zones.

- Clustering: Algorithms such as DBSCAN, ST-DBSCAN, and K-Means will be applied to incident location data to detect clusters of unsafe activity [9], [10]. This allows the identification of “hotspots” such as unsafe bus stops, routes, or school vicinities.
- Time-of-Day Analysis: Historical patterns will be combined with clustering results to predict time-sensitive risks. For example, certain locations may be classified as low risk in the morning but high risk in the evening. This approach addresses survey findings where

participants identified daytime risks (morning and afternoon) as equally unsafe as night-time conditions.

- Heatmaps: A GIS-based visualization layer will display unsafe zones in the mobile app, enabling users to make informed travel decisions.

This methodology is built on prior work by Jain and Kumar & Kaur, who demonstrated the power of spatiotemporal clustering in predicting crime patterns [9], [10]. By integrating time and place, the system provides dynamic and context-aware risk predictions.

3.1.3 Crowdsourced Safety Ratings

The system also introduces a community-driven Safety Rating Board to complement AI-driven predictions. Users will be able to provide safety ratings for public transport, streets, schools, and community institutions.

To ensure reliability and trustworthiness, the ratings will undergo AI-based validation mechanisms including:

- Trust-score weighting: Scores will be adjusted based on user history, consistency, and behavioral footprints, similar to methods proposed by Rathi & Mishra [7].
- Anomaly detection: AI will flag sudden spikes in ratings or coordinated spam campaigns, as described by Ahmed [13].
- Reputation scoring: Frequent, reliable contributors will have higher weighting compared to new or unverified users.

This ensures that fake or malicious inputs do not distort the safety ratings, providing communities with credible and actionable safety insights.

3.1.4 Alert and Guidance Delivery

The end-user functionality of the system involves generating alerts and safety tips in Sinhala and English. This component includes:

- Real-time push notifications: Delivered when users enter or approach identified unsafe zones, e.g., “Warning: High reports of harassment near this bus route between 6–8 PM.”
- Preventive guidance messages: Based on risk factors such as alcohol-related hotspots or poorly lit areas, e.g., “Avoid walking alone along X Street after 9 PM.”
- Child-focused alerts: Special features for parents and guardians, including alerts about unsafe school transport zones or abduction-prone areas.

- General awareness tips: Safety reminders, legal rights information, and emergency contacts provided in both Sinhala and English.

This feature directly addresses survey findings where 90% of respondents agreed that predictive safety alerts would reduce risks. By delivering alerts in multiple languages, the system ensures wide accessibility and cultural inclusiveness.

3.2 System Architecture Diagram

Inside of the component there are several tasks which are divided for the purpose of understanding and to do further the implementations.

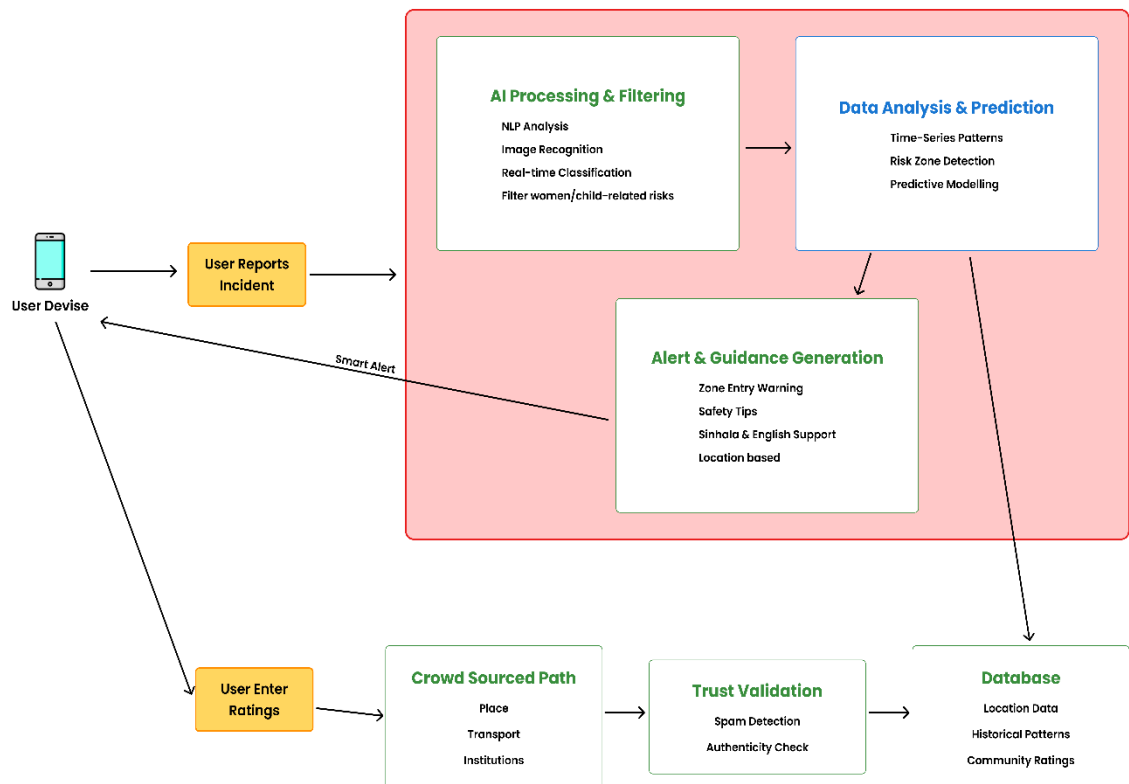


Figure 3. 3: High Level Architecture Diagram

The proposed system architecture follows a modular design, where each component performs a specialized role to ensure a smooth workflow from data collection to safety prediction and alert delivery. At the front end, the mobile application provides the main interface for users

to interact with the system. Through the app, users can easily submit incident reports in text, audio, or multimedia format, while also receiving safety alerts and visualizing heatmaps of unsafe areas. Once submitted, these reports are handled by the report collection layer, which anonymizes sensitive information, normalizes input data, and prepares it for downstream processing. This ensures that reports are both privacy-preserving and structured for accurate AI-based analysis.

The processed reports are then passed into the AI filters and classifiers, which form the core intelligence of the system. Using a bilingual Sinhala-English NLP pipeline, the system analyzes incoming data and identifies women- and child-related risks such as harassment, unsafe transport, domestic abuse, or child endangerment. By applying advanced classification techniques, this module ensures that only relevant incidents are flagged for prediction and visualization. These validated reports are then fed into the risk predictor, which combines geospatial clustering algorithms with time-series analysis to identify risk-prone areas and forecast unsafe times. The output is visualized as dynamic heatmaps, highlighting emerging hotspots and providing preventive insights.

Parallel to this, the system integrates a crowdsourced safety database, where users can provide ratings for public transport, institutions, and local areas. To ensure reliability, these ratings undergo trust-score validation and spam filtering before being incorporated into the main risk database. The combined insights from AI classification, predictive modeling, and community input are made available through a police dashboard and moderator panel, enabling law enforcement and moderators to view validated incidents, predictive hotspots, and rating summaries for decision-making. Finally, the alert engine generates personalized push notifications and bilingual safety tips, guiding users in Sinhala and English. This modular architecture not only ensures scalability and resilience but also facilitates seamless integration with law enforcement while empowering communities to take an active role in improving safety for women and children.

3.3 Project Requirements

3.3.1 Functional Requirements and Non-Functional Requirements

Functional Requirements

- The system must automatically filter and classify incoming SOS reports and incident submissions.
- The system must generate dynamic geospatial heatmaps that visualize unsafe areas.
- Predictive clustering must highlight time-sensitive risk patterns (e.g., unsafe routes at specific times).
- The system must allow users to provide ratings for public transport, streets, and institutions.
- The system must deliver real-time push notifications and safety guidance in Sinhala and English.

Non-Functional Requirements

- Accuracy - Incident classification and predictive models must achieve $\geq 85\%$ accuracy in detecting women/child-related risks.
- Reliability and Availability - The system must provide up time for core features and ensure uninterrupted alert delivery.
- Performance and Latency - Push alerts must be delivered within 3 seconds of risk detection.
Incident classification should be completed within 1 second per report.
- Security and Privacy - All sensitive data (location, media, personal details) must be encrypted at rest and in transit. Reports must allow anonymous submissions to protect victims.
- Scalability - The system must support increasing user loads (10,000+ daily users) without performance degradation.

3.3.2 User Requirements

- Users must be able to view unsafe areas and receive predictive safety alerts before entering high-risk zones.
- Users must be able to contribute to safety ratings for places and transport services.
- The system must ensure anonymity, protect personal data, and provide users with control over what information is shared.
- The interface must fully support Sinhala and English to remove language barriers in reporting and receiving alerts.

3.3.3 System Requirements

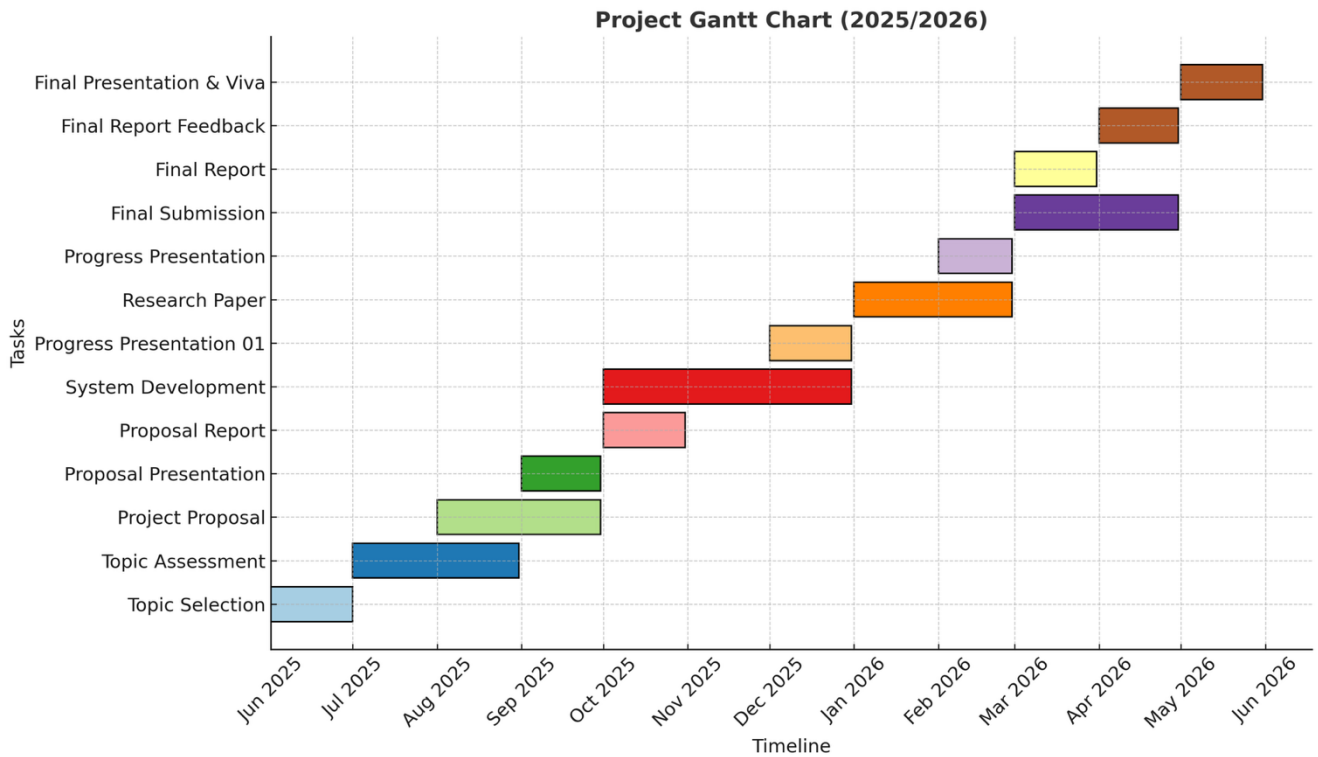
- Software requirements
- Flutter
- Python
- FastAPI
- React / React Native
- TensorFlow
- Google Map
- MySQL / MongoDB
- AWS

3.3.4 Personnel Requirements

Collecting resources and dataset of diseases.

- Sri Lanka Police
- Moderators

4. GANTT CHART



5. BUDGET AND BUDGET JUSTIFICATION

Table 6. 1: Budget Plan

Requirement	Cost (Rs.)
Travelling Cost for data collection	6000.00
vps	7500
Mobile app hosting on Play Store	8200.00
Mobile app hosting on App Store	20000
Total	39700

6. COMMERCIALIZATION

6.1 Target Audience and Market Space.

- **Target Audience**
Women
Parents
school authorities
Public transport users.
Researchers and Citrus Research Institute
- **Market Space**
No need for advanced knowledge in technology.
No age limitation for the users.

REFERENCES

- [1] V. K. Jewani, P. E. Ajmire, S. A. Chaurasia, and G. Brijwani, “Artificial Intelligence: A Smart and Empowering Approach to Women’s Safety,” in *Impact of AI on Advancing Women’s Safety*, Hershey, PA, USA: IGI Global, Dec. 2023.
- [2] S. Ponnusamy, *et al.*, *AI Tools and Applications for Women’s Safety*. Hershey, PA, USA: IGI Global, 2024.
- [3] P. K. Yu, “The Algorithmic Divide and Equality in the Age of Artificial Intelligence,” *Florida Law Review*, 2020.
- [4] *Computing-Based Applications for Women’s Safety*. London, UK: Taylor & Francis, 2024.
- [5] K. Kanimozhi and K. Arputharaj, “Smart Surveillance System for Women and Child Safety using Deep Learning,” *Journal of Ambient Intelligence and Humanized Computing*, 2022.
- [6] A. Thakare and G. Shinde, “Women Safety Device Using AI and Machine Learning,” *Procedia Computer Science*, 2023.

- [7] V. Rathi and A. Mishra, "Crowdsourcing Safety: Rating System Design Against Misuse," *IEEE Access*, 2021.
- [8] H. Perera, *et al.*, "Sinhala-English Code-Mixed Language Identification and NLP Models," in *Proc. Int. Conf. Natural Language Processing (ICON)*, 2020.
- [9] A. Jain, *et al.*, "Crime Pattern Analysis Using Spatiotemporal Clustering and Machine Learning," *Int. J. Data Science*, 2019.
- [10] P. Kumar and M. Kaur, "Real-time Crime Mapping and Prediction Using Geospatial AI," *Journal of Location Intelligence*, 2021.
- [11] S. Sarma and P. Prakash, "Early Detection of Domestic Violence via Social Media Using NLP," *Social Network Analysis and Mining*, 2022.
- [12] D. Birant and A. Kut, "ST-DBSCAN: An Algorithm for Clustering Spatial–Temporal Data," *Data & Knowledge Engineering*, 2007.
- [13] S. Ahmed, *et al.*, "Crowdsourcing Safety in Developing Countries: Trust-Weighted Systems," *Information Systems Frontiers*, 2020.
- [14] R. Patel and S. Verma, "AI-powered Geo-tagged Safety Alert System for Urban Women," *Urban Informatics Journal*, 2022.
- [15] H. Weerasinghe and A. Jayasuriya, "Sinhala-English Multilingual NLP Toolkit for Social Safety Apps," in *Proc. Colombo NLP Workshop*, Colombo, Sri Lanka, 2023.
- [16] N. Goyal and A. Thakur, "Women Safety Using AI-Based Smart Wearables with Location Prediction," *Procedia Computer Science*, 2023.
- [17] M. Rani and R. Kapoor, "Smart Safety App for Women with AI Prediction Models," *Int. J. Engineering Research*, 2019.
- [18] K. Joshi and R. Mishra, "Analyzing Patterns of Child Endangerment Reports Using Machine Learning," *Child Abuse & Neglect Journal*, 2020.
- [19] M. Ibrahim, *et al.*, "Addressing Multilingual Challenges in NLP for Public Safety Apps," *Transactions in Asian Language Processing*, 2021.

APPENDICES

Survey link: <https://docs.google.com/forms/d/e/1FAIpQLScbQR9I3wusIZ60APv3B-53PG5xOts8eiUWWa8vr4FF1iC1Ww/viewform?usp=dialog>